

PRAKTIKUM SISTEM OPERASI

MODUL 4

MEMBACA SOURCE CODE XINU



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

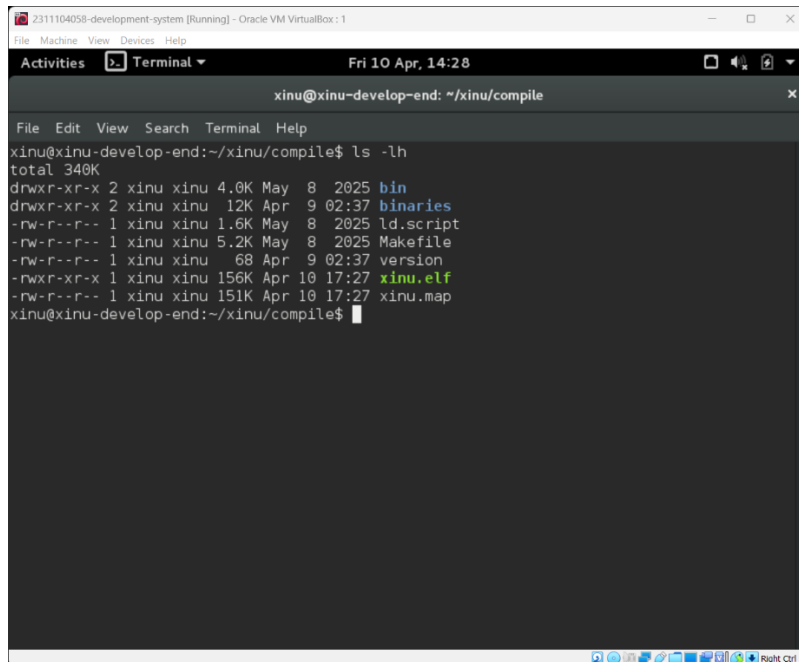
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

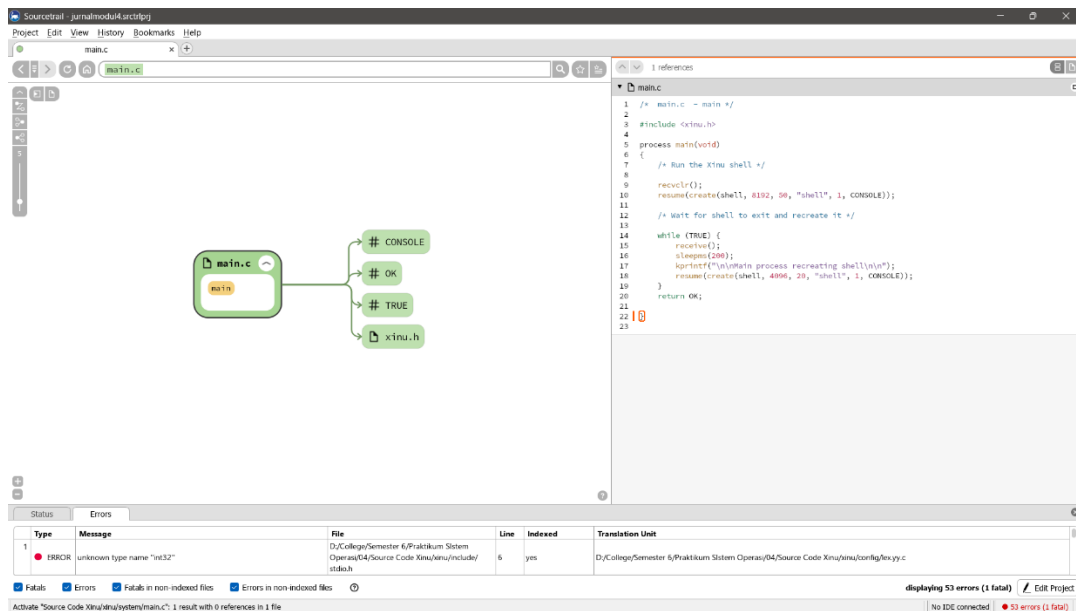
JURNAL

1. Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?
File executable atau image sistem operasi hasil kompilasi bernama xinu dengan ukuran 156k dan ada di folder xinu/compile.



```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu/compile$ ls -lh
total 340K
drwxr-xr-x 2 xinu xinu 4.0K May  8 2025 bin
drwxr-xr-x 2 xinu xinu 12K Apr  9 02:37 binaries
-rw-r--r-- 1 xinu xinu 1.6K May  8 2025 ld.script
-rw-r--r-- 1 xinu xinu 5.2K May  8 2025 Makefile
-rw-r--r-- 1 xinu xinu  68 Apr  9 02:37 version
-rwxr-xr-x 1 xinu xinu 156K Apr 10 17:27 xinu.elf
-rw-r--r-- 1 xinu xinu 151K Apr 10 17:27 xinu.map
xinu@xinu-develop-end:~/xinu/compile$
```

2. Membaca source code Xinu

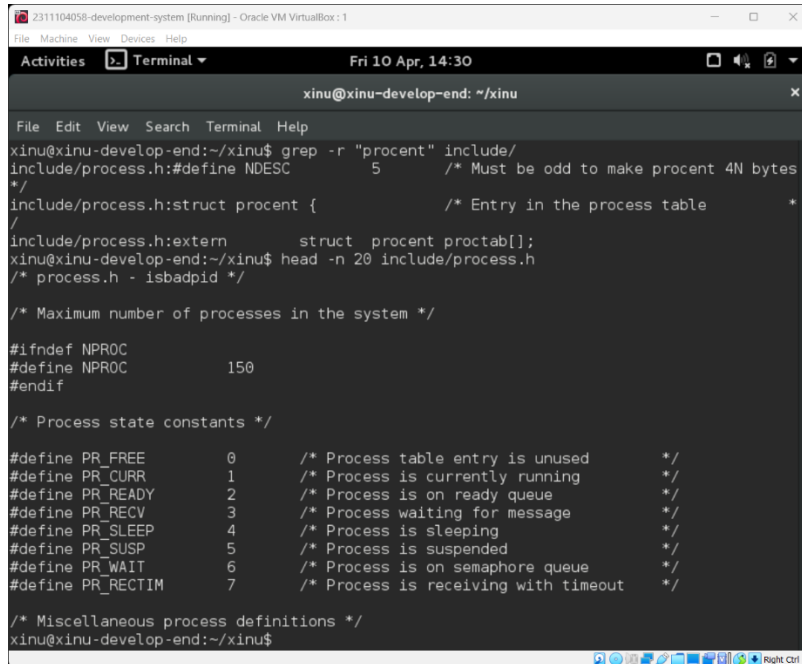


The screenshot shows the SourceTrail IDE with the project 'jurnalmodul4.kurtlog' open. The file 'main.c' is selected, and its source code is displayed in the editor. The code includes a comment '/* Run the xinu shell */' and a function 'process_main(void)' that creates a shell process. The IDE also shows a diagram of the code structure with nodes for 'main.c', 'main', 'CONSOLE', 'OK', 'TRUE', and 'xinu.h'. The status bar at the bottom indicates 'displaying 53 errors (1 fatal)' and '53 errors (1 fatal)'.

```
1 /* main.c - main */
2
3 #include <xinu.h>
4
5 process_main(void)
6 {
7     /* Run the xinu shell */
8
9     recvr();
10    resume(create(shell, 8192, 30, "shell", 1, CONSOLE));
11
12    /* Wait for shell to exit and recreate it */
13
14    while (TRUE) {
15        receive();
16        sleepms(200);
17        kprintf("in main process recreating shell\n");
18        resume(create(shell, 4096, 30, "shell", 1, CONSOLE));
19    }
20    return OK;
21 }
22
23
```

3. Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?

Struktur data proses pada Xinu OS disimpan di dalam file berekstensi .h bernama process.h yang terletak di dalam direktori include/. Di dalam struktur data, tersimpan berbagai informasi penting (Process Control Block) untuk manajemen proses seperti State atau status proses, Prioritas proses (scheduling priority), Alamat dan ukuran memori stack dari proses, Nama proses, dan Process ID dari parent yang membuat proses tersebut.



```
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu$ grep -r "procent" include/
include/process.h:#define NDESC      5      /* Must be odd to make procent 4N bytes
*/
include/process.h:struct procent {          /* Entry in the process table */
/
include/process.h:extern      struct procent proctab[];
xinu@xinu-develop-end:~/xinu$ head -n 20 include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      150
#endif

/* Process state constants */

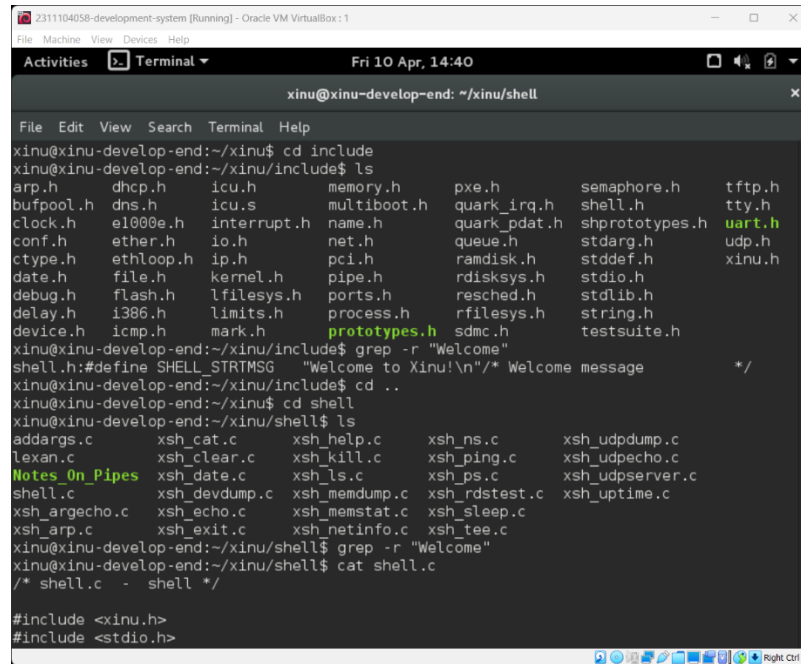
#define PR_FREE      0      /* Process table entry is unused */
#define PR_CURR      1      /* Process is currently running */
#define PR_READY     2      /* Process is on ready queue */
#define PR_RECV      3      /* Process waiting for message */
#define PR_SLEEP     4      /* Process is sleeping */
#define PR_SUSP      5      /* Process is suspended */
#define PR_WAIT      6      /* Process is on semaphore queue */
#define PR_RECTIM    7      /* Process is receiving with timeout */

/* Miscellaneous process definitions */
xinu@xinu-develop-end:~/xinu$
```

4. Mengubah welcome banner pada Xinu

a. Carilah file yang menyimpan banner Xinu!

File header yang mendefinisikan/menyimpan teks banner Xinu adalah shell.h yang berada pada direktori xinu/include/.



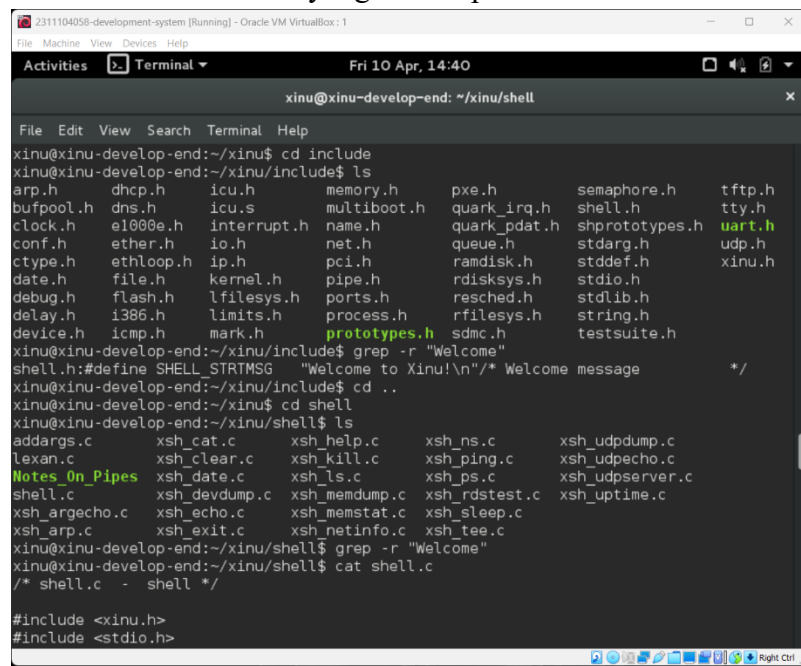
```
xinu@xinu-develop-end:~/xinu$ cd include
xinu@xinu-develop-end:~/xinu/include$ ls
arp.h      dhcp.h      icu.h      memory.h   pxe.h      semaphore.h  tftp.h
bufpool.h  dns.h      icu.s      multiboot.h  quark_irq.h  shell.h      tty.h
clock.h    el000e.h  interrupt.h  name.h      quark_pdat.h  shprototypes.h  uart.h
conf.h     ether.h    io.h      net.h      queue.h      stdarg.h      udp.h
ctype.h    ethloop.h ip.h      pci.h      ramdisk.h    stddef.h      xinu.h
date.h     file.h    kernel.h  pipe.h      rdisksys.h   stdio.h
debug.h    flash.h   lfilesys.h  ports.h    resched.h    stdlib.h
delay.h    i386.h   limits.h   process.h   rfilesys.h   string.h
device.h   icmp.h   mark.h     prototypes.h  sdmc.h      testsuite.h

xinu@xinu-develop-end:~/xinu/include$ grep -r "Welcome"
shell.h:#define SHELL_STRMSG  "Welcome to Xinu!\n"/* Welcome message      */
xinu@xinu-develop-end:~/xinu/include$ cd ..
xinu@xinu-develop-end:~/xinu$ cd shell
xinu@xinu-develop-end:~/xinu/shell$ ls
addargs.c  xsh_cat.c  xsh_help.c  xsh_ns.c  xsh_udpdump.c
lexan.c    xsh_clear.c  xsh_kill.c  xsh_ping.c  xsh_udpecho.c
Notes_On_Pipes  xsh_date.c  xsh_ls.c  xsh_ps.c  xsh_udpserver.c
shell.c    xsh_devdump.c  xsh_mempdump.c  xsh_rdstest.c  xsh_uptime.c
xsh_argecho.c  xsh_echo.c  xsh_memstat.c  xsh_sleep.c
xsh_arp.c      xsh_exit.c  xsh_netinfo.c  xsh_tee.c
xinu@xinu-develop-end:~/xinu/shell$ grep -r "Welcome"
xinu@xinu-develop-end:~/xinu/shell$ cat shell.c
/* shell.c - shell */

#include <xinu.h>
#include <stdio.h>
```

b. Carilah file yang menampilkan banner Xinu!

File C yang bertugas memanggil dan mencetak banner tersebut ke layar konsol terminal adalah shell.c yang berada pada direktori xinu/shell/.



```
xinu@xinu-develop-end:~/xinu$ cd include
xinu@xinu-develop-end:~/xinu/include$ ls
arp.h      dhcp.h      icu.h      memory.h   pxe.h      semaphore.h  tftp.h
bufpool.h  dns.h      icu.s      multiboot.h  quark_irq.h  shell.h      tty.h
clock.h    el000e.h  interrupt.h  name.h      quark_pdat.h  shprototypes.h  uart.h
conf.h     ether.h    io.h      net.h      queue.h      stdarg.h      udp.h
ctype.h    ethloop.h ip.h      pci.h      ramdisk.h    stddef.h      xinu.h
date.h     file.h    kernel.h  pipe.h      rdisksys.h   stdio.h
debug.h    flash.h   lfilesys.h  ports.h    resched.h    stdlib.h
delay.h    i386.h   limits.h   process.h   rfilesys.h   string.h
device.h   icmp.h   mark.h     prototypes.h  sdmc.h      testsuite.h

xinu@xinu-develop-end:~/xinu/include$ grep -r "Welcome"
shell.h:#define SHELL_STRMSG  "Welcome to Xinu!\n"/* Welcome message      */
xinu@xinu-develop-end:~/xinu/include$ cd ..
xinu@xinu-develop-end:~/xinu$ cd shell
xinu@xinu-develop-end:~/xinu/shell$ ls
addargs.c  xsh_cat.c  xsh_help.c  xsh_ns.c  xsh_udpdump.c
lexan.c    xsh_clear.c  xsh_kill.c  xsh_ping.c  xsh_udpecho.c
Notes_On_Pipes  xsh_date.c  xsh_ls.c  xsh_ps.c  xsh_udpserver.c
shell.c    xsh_devdump.c  xsh_mempdump.c  xsh_rdstest.c  xsh_uptime.c
xsh_argecho.c  xsh_echo.c  xsh_memstat.c  xsh_sleep.c
xsh_arp.c      xsh_exit.c  xsh_netinfo.c  xsh_tee.c
xinu@xinu-develop-end:~/xinu/shell$ grep -r "Welcome"
xinu@xinu-develop-end:~/xinu/shell$ cat shell.c
/* shell.c - shell */

#include <xinu.h>
#include <stdio.h>
```

- Untuk mengubah welcome banner bisa dilakukan pada file shell.h yang berada pada folder include atau dilakukan di file shell.c di folder shell.

```
2311104058-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Fri 10 Apr, 14:50 xinu@xinu-develop-end: ~/xinu/include

GNU nano 2.2.6 File: shell.h Modified

/* Shell banner (assumes VT100) */

#define SHELL_BAN0 "\033[31;1m"
#define SHELL_BAN1 "-----Dimastian-----"
#define SHELL_BAN2 " "
#define SHELL_BAN3 "  \\\\/  /  _  _  \\\\/  |  |  |  |  |  "
#define SHELL_BAN4 "  \\\\/  /  _  _  \\\\/  |  |  |  |  |  "
#define SHELL_BAN5 "  \\\\/  /  _  _  \\\\/  |  |  |  |  |  "
#define SHELL_BAN6 "  \\\\/  /  _  _  \\\\/  |  |  |  |  |  "
#define SHELL_BAN7 "  \\\\/  /  _  _  \\\\/  |  |  |  |  |  "
#define SHELL_BAN8 "-----2311104058-----"
#define SHELL_BAN9 "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT "xsh $ " /* Command prompt */
#define SHELL_STRMSG "Welcome to XINU!\n" /* Welcome message */
#define SHELL_EXITMSG "Shell closed\n" /* Shell exit message */
#define SHELL_SYNERMSG "Syntax error\n" /* Syntax error message */
#define SHELL_CREATMSG "Cannot create process\n" /* command error */
#define SHELL_INERRMSG "Cannot open file %s for input\n" /* Input err */
#define SHELL_OUTERRMSG "Cannot open file %s for output\n" /* Output err */

Get Help WriteOut Read File Prev Page Cut Text Cur Pos
Exit Justify Where Is Next Page UnCut Text To Spell

2311104058-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Fri 10 Apr, 15:01 xinu@xinu-develop-end: ~/xinu/shell

GNU nano 2.2.6 File: shell.c Modified

/* by cmdlookup */
did32 pipedev; /* Device ID of a pipe device */
int32 i; /* Index into array of tokens */
char *p; /* Pointer to cmd name */
bool8 err; /* Did an error occur? */
int32 msg; /* Message from receive() for child termination */
int32 tmparg; /* Temporary address used when creating a child process; later replaced by addargs */

/* Print shell banner and startup message */

kprintf("Hello World!\n");
kprintf("I'm the first Xinu app and running function main() in system/main.c.\n$");
kprintf("My creator will turn itself into the do-nothing null process.\n");
kprintf("I'll create a second XINU app that runs shell() in shell/shell.c as$");
kprintf("You can do something else or do nothing; it's completely up to you.\n$");
kprintf("...creating a shell\n");
kprintf("XINU\n");
fprintf(dev, "\n\n%s\n\n%s\n\n%s\n\n%s\n\n%s\n\n%s\n\n",
SHELL_BAN0,SHELL_BAN1,SHELL_BAN2,SHELL_BAN3,SHELL_BAN4,
```

d. Compile

The screenshot shows a virtual machine window titled "z311104058-development-system [Running] - Oracle VM VirtualBox - 1". Inside the VM, a terminal window titled "xinu@xinu-develop-end: ~/xinu/compile" is open. The terminal output shows the XINU boot process, including the IP address 192.168.201.115, a "Hello World!" message, and a large ASCII art logo for "XINU". The terminal also displays the text "Welcome to Xinu!" and a prompt "xsh \$".

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
Obtained IP address 192.168.201.115 (0xc0a8c973)

Hello World!
Im the first Xinu app and running function main() in system/main.c.
I was created by nulluser() using create().
My creator will turn itself into the do-nothing null process.
I will create a second XINU app that runs shell() in shell/shell.c as an example.
You can do something else or do nothing; it's completely up to you.
...creating a shell
XINU

-----Dimastian-----
XINU
-----2311104058-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```